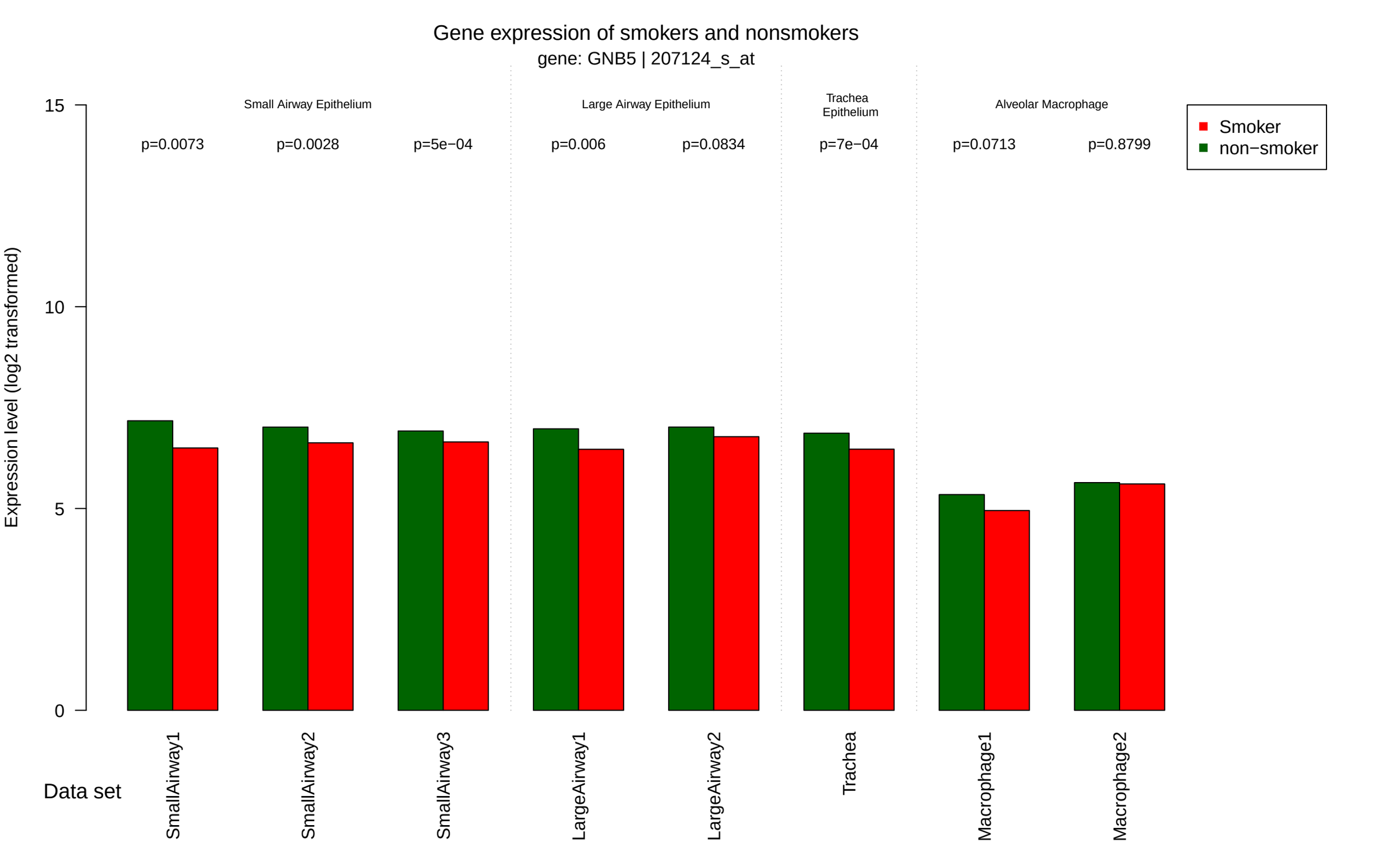
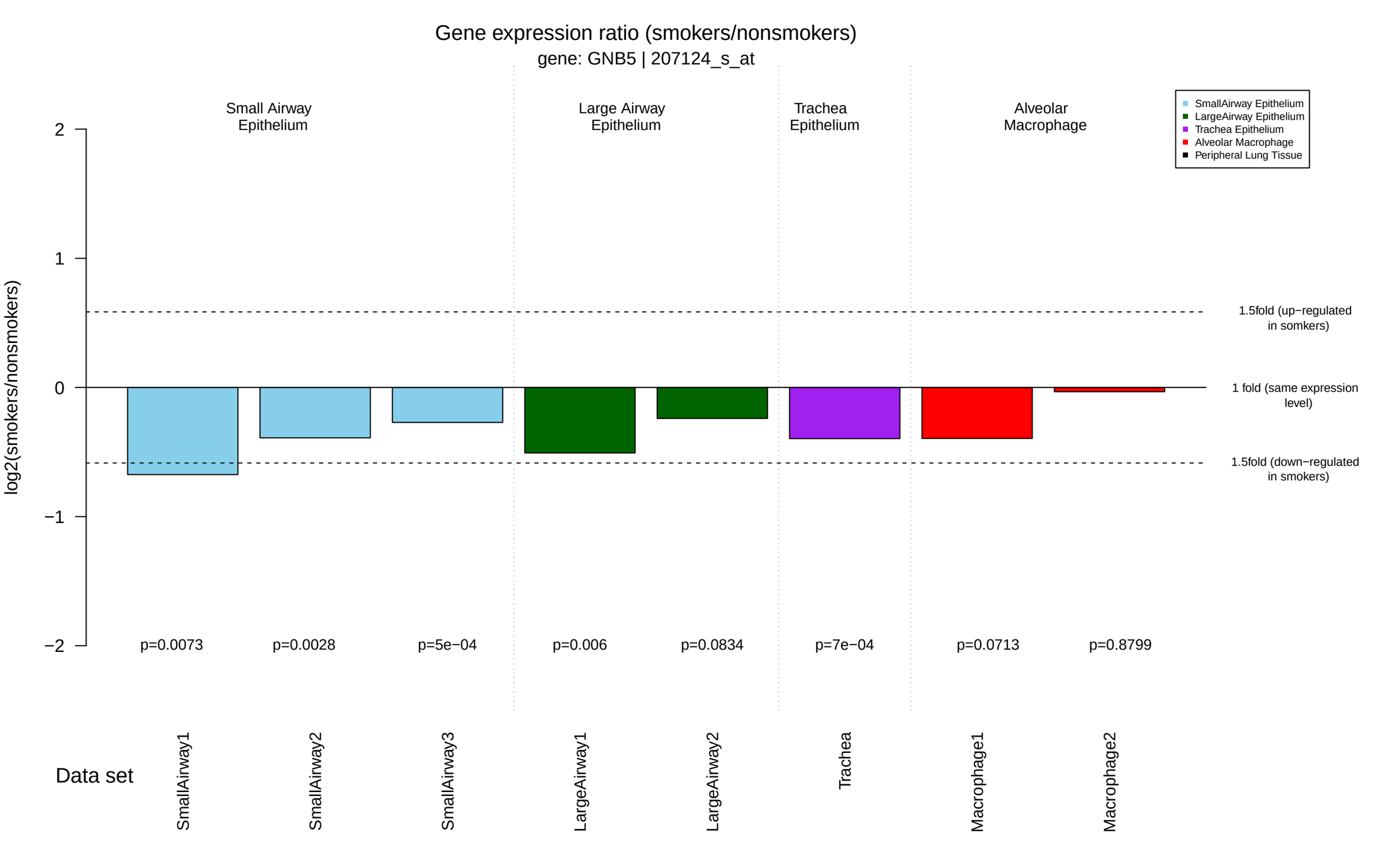
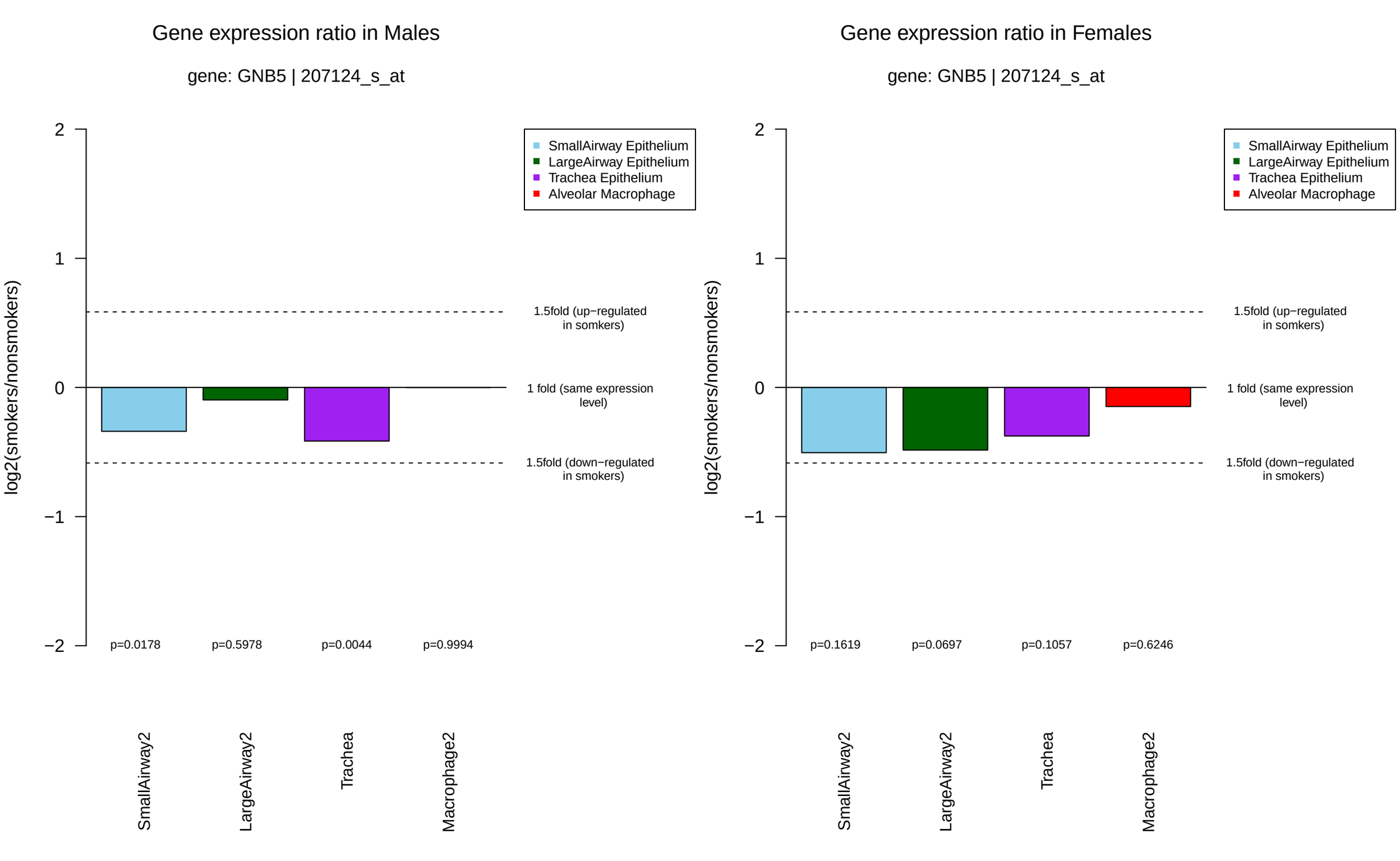
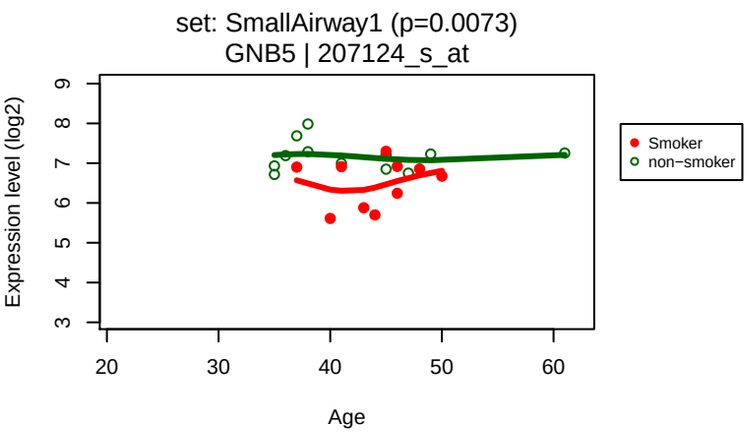
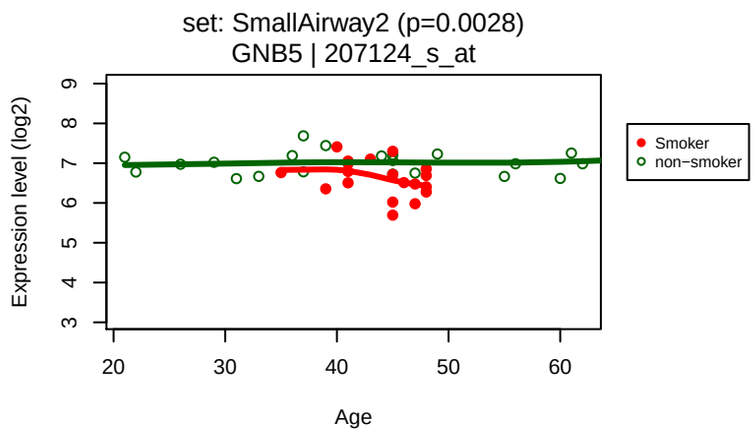
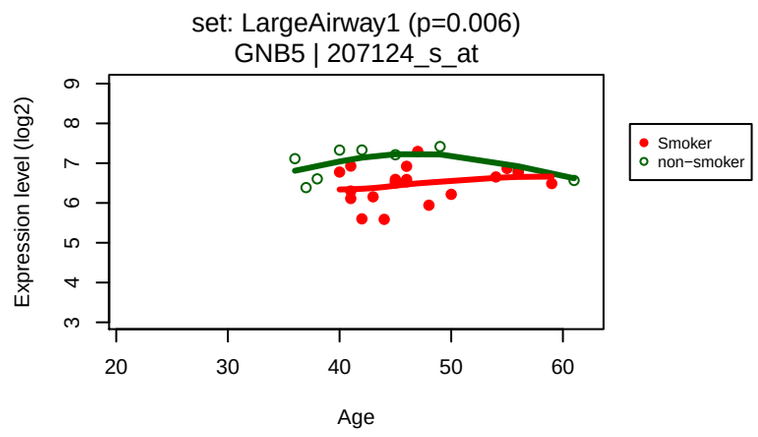

Gene expression of smokers and nonsmokers
gene: GNB5 | 207124_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0073
p=0.0028
p=5e−04
p=0.006
p=0.0834
p=7e−04
p=0.0713
p=0.8799
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

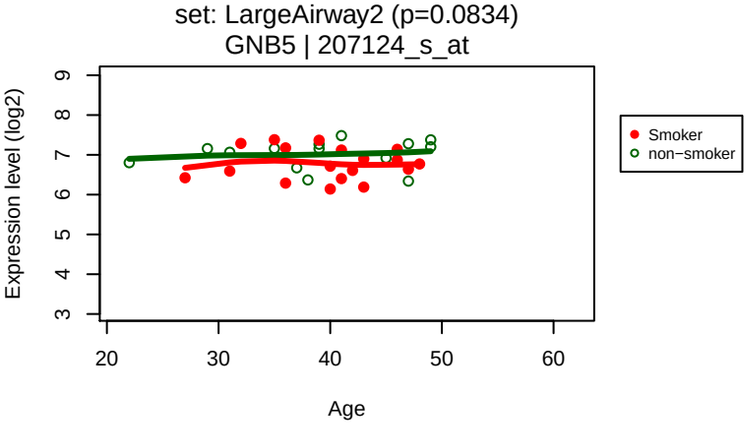

Gene expression ratio (smokers/nonsmokers)
gene: GNB5 | 207124_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0073
p=0.0028
p=5e−04
p=0.006
p=0.0834
p=7e−04
p=0.0713
p=0.8799
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

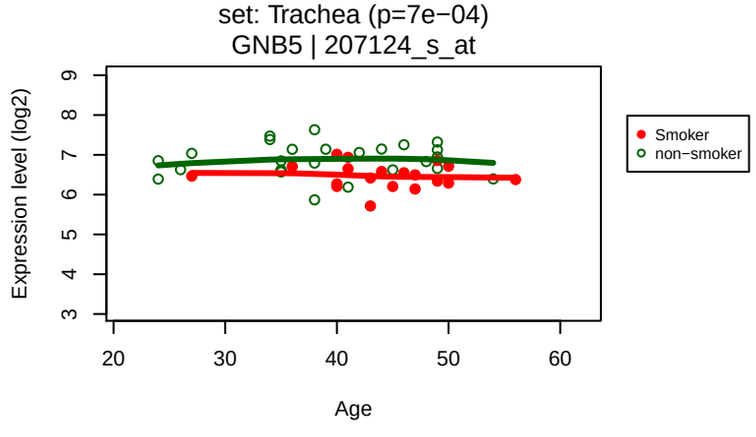

Gene expression ratio in Males
gene: GNB5 | 207124_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0178
p=0.5978
p=0.0044
p=0.9994
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNB5 | 207124_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1619
p=0.0697
p=0.1057
p=0.6246
SmallAirway2
LargeAirway2
Trachea
Macrophage2

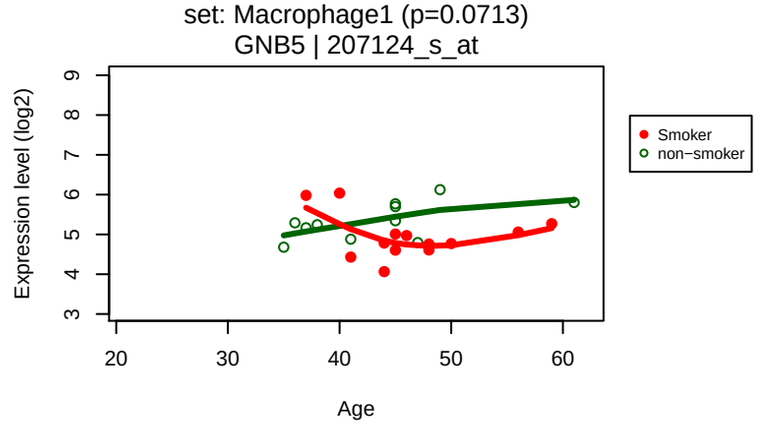

set: SmallAirway1 (p=0.0073)
GNB5 | 207124_s_at
Expression level (log2)
Age
Smoker
non−smoker

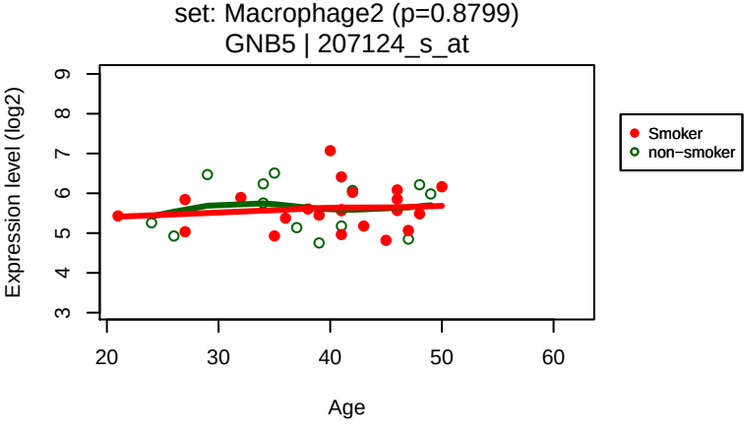

set: SmallAirway2 (p=0.0028)
GNB5 | 207124_s_at
Expression level (log2)
Age
Smoker
non−smoker

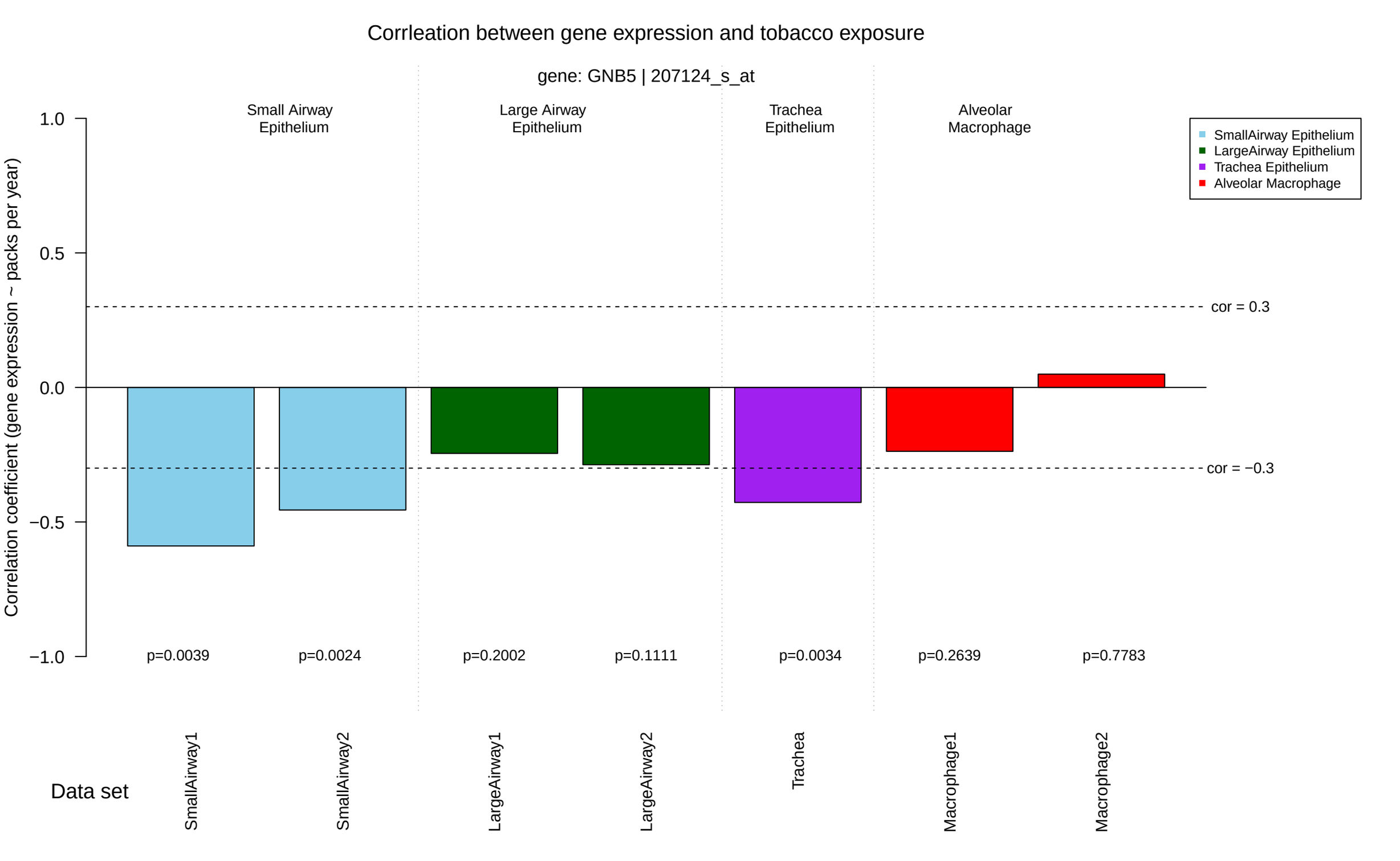

set: LargeAirway1 (p=0.006)
GNB5 | 207124_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0834)
GNB5 | 207124_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=7e−04)
GNB5 | 207124_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0713)
GNB5 | 207124_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8799)
GNB5 | 207124_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNB5 | 207124_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0039
p=0.0024
p=0.2002
p=0.1111
p=0.0034
p=0.2639
p=0.7783
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2